



getaddrinfo()

DNS and service resolution. It handles:

- Hostname resolution (example.com → IP address)
- Service name to port ("http" → 80)
- IPv4/IPv6 abstraction

```
#include <sys/socket.h>
#include <netdb.h>

struct addrinfo hints;
struct addrinfo *result;
struct addrinfo *rp;
int sock;

std::memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_UNSPEC;      // IPv4 or IPv6
hints.ai_socktype = SOCK_STREAM; // TCP
hints.ai_flags = AI_PASSIVE;     // For bind (wildcard addresses)

// Get address info for port 8080
int ret = getaddrinfo(NULL, "8080", &hints, &result);
if (ret != 0)
{
    std::cerr << "getaddrinfo error: " << gai_strerror(ret) <
```

```

< std::endl;
    return -1;
}

// Try each address until bind succeeds
for (rp = result; rp != NULL; rp = rp->ai_next)
{
    sock = socket(rp->ai_family, rp->ai_socktype, rp->ai_protocol);
    if (sock < 0)
        continue;

    if (bind(sock, rp->ai_addr, rp->ai_addrlen) == 0)
        break; // Success

    close(sock);
}

freeaddrinfo(result); // IMPORTANT: Free the linked list

if (rp == NULL)
    return -1; // No address worked

```

<https://www.man7.org/linux/man-pages/man3/getaddrinfo.3.html>